

# DSGE Models

Pavel Solís

CIDE

2026

# Introduction

- Macroeconomists study both long-run and short-run topics
  - Long-run: Economic growth, natural rate of unemployment
  - Short-run: Business cycle, stabilization (monetary, fiscal) policy
- We will focus on dynamic models of aggregate demand and supply to **quantitatively** analyze the business cycle and stabilization policy
  - Nominal vs real variables
  - Closed vs open economy
  - Finite vs infinite horizon
  - Continuous vs discrete time
  - Perfect foresight vs uncertainty
  - Endogenous vs exogenous variables

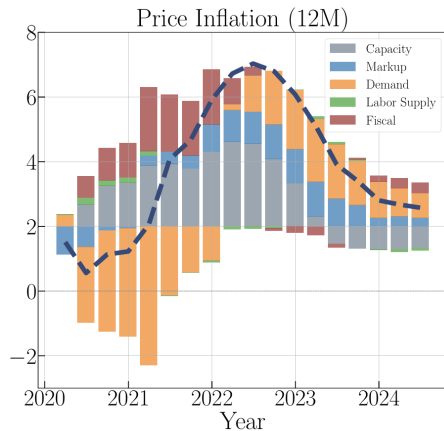
# DSGE Models

- Mathematical representations of an economy subject to shocks
- Acronym stands for Dynamic Stochastic General Equilibrium
  - Dynamic: Agents make intertemporal decisions (today vs future)
  - Stochastic: Random exogenous shocks capture uncertainty
  - GE: All markets interact and prices adjust so that supply equals demand
- DSGE models are a central tool for macroeconomic analysis
- Used by central banks and international organizations (eg IMF, WB)
  - Simulate and forecast variables of interest
  - Scenario and counterfactual analysis

## Structural vs Reduced-Form Models

- DSGE models force us to make structure of economy explicit
  - We model households, firms, frictions and policy rules
  - Microeconomic foundations: Optimizing agents (utility, profits)
  - They are internally consistent, not subject to Lucas critique (1976)
- With explicit structure, mechanisms can be isolated, quantified, tested
  - We can trace evolution of a variable back to specific mechanisms
- DSGEs are thus useful to answer structural questions
  - Example: Main driver behind post-pandemic inflation in US? Supply bottlenecks? Demand pressure? Pricing power? Policy response?

# Example



Alves and Violante, 2026. Inflation vs Inclusion: Stabilization Policy in the Wake of the Pandemic, CEPR Discussion Paper.

# Module Outline

- We will cover the canonical models
  - Real Business Cycle (RBC) - flexible prices without money
  - Money in the Utility Function (MIUF) - flexible prices with money
  - New Keynesian (NK) - price frictions
- RBC model set standard for analysis of economic fluctuations
  - Kydland and Prescott (1982)
  - Productivity shocks, perfect competition, no role for monetary policy
- Models then introduced new shocks/frictions from Keynesian principles
- We will only cover price frictions; recent models add more
  - Habit formation, investment adjustment costs, financial, etc.

# Structure of Models

- Assumptions
- Agents' optimization problems (Lagrange's method)
- Model's equilibrium (short-run)
- Steady state (long-run)
- Calibration
- Log-linearization (Uhlig's method)
- Dynare implementation
- Impulse response functions (IRFs)
  - How endogenous variables respond to exogenous shocks

# Representative Agents

- Agents in economy are heterogeneous
  - Agents: Households, firms, financial intermediaries, foreigners
  - Consumers have different preferences, firms use different technologies
- Any economic model is a simplified description of reality
- Canonical DSGE models make use of **representative agents**
  - Goal: Assess macroeconomic policy by analyzing their interactions
  - Recent models use heterogeneous agents explicitly (e.g., HANK models)
- **Simplifying assumption:**
  - Large number of identical agents (firms, families) make decisions with infinite time horizons



# CRRA

- $U(C)$  denotes utility of consumption with  $U'(C) > 0$  and  $U''(C) < 0$
- Relative risk aversion (RRA) is defined as follows

$$RRA = -C \frac{U''(C)}{U'(C)}$$

- A utility function with a constant RRA (CRRA) is of the form

$$U(C) = \frac{C_t^{1-\gamma}}{1-\gamma}, \quad \gamma \geq 0, \gamma \neq 1; \quad U(C) = \ln(C), \quad \gamma = 1$$

- Verify  $RRA = \gamma$  (i.e., same RRA irrespective of  $C$  level)
- Empirical studies based on actual behavior of individuals find  $\gamma \in [1, 4]$
- Individuals with higher RRAs are more averse to risk

# Calibration

- Dynamic macroeconomic models are a set of **parameters** and **equations** describing relationships between macroeconomic variables
  - Consumption, investment, output, trade balance (exports - imports), employment, prices, interest rates, exchange rates
- Parameter values are key inputs to numerically implement a model
  - Estimation: Using actual data
  - Calibration: Assign values based on consensus and/or related papers
    - Proceed with caution and common sense
- **Important:** Match length of time period (quarterly, annually) in model

# Dynare

- Free collection of routines for solving DSGE (and OLG) models
  - Rational expectations, perfect foresight, learning
  - Discrete time models only (perturbation techniques around a SS)
- Project founded by Michel Juillard in 1994; version 7.0 (March 2026)
- Widely used by academic and policy institutions
- A Matlab license is required to run Dynare
  - Alternatives: Octave (limited) and Julia (beta version)
- In practice:
  - Write text (.mod) file: variables, parameters, equations, shocks, output
  - Dynare's preprocessor translates the .mod file into Matlab code
  - Matlab numerically solves model and displays results (e.g. IRFs)